

TCR–Epitope Binding Prediction

Baseline Results Summary

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1. Project Overview

This document summarises the baseline machine learning pipeline built to predict whether a T-cell receptor (TCR) binds a given epitope. The goal is to establish reference metrics before moving to deep learning and graph-based approaches.

1.1. Data

- **Source:** iggytop database ([biocypher/iggytop](#))
- **Size:** ~300,000 positive TCR–epitope pairs extracted via `scirpy.get.airr_context`
- **Sequences:** CDR3 β (VDJ₁ / TRB) and epitope amino-acid strings
- **Negatives:** random mismatched pairing (each TCR paired with a random epitope it did not come from)
- **Final dataset:** ~600,000 samples (1:1 positive:negative ratio)

1.2. Feature Engineering

No hand-crafted features. Both sequences were embedded using **ESM2** ([facebook/esm2_t6_8M_UR50D](#)), a protein language model pre-trained on 250M protein sequences, yielding a 320-dimensional vector per sequence. TCR and epitope embeddings were concatenated into a **640-dimensional feature vector**.

1.3. Models

- **Logistic Regression** — linear baseline (scikit-learn, saga solver, balanced class weights)
- **Random Forest** — non-linear baseline (100 trees, max depth 15, balanced class weights)

2. Model Performance

2.1. Interpretation

- Random Forest substantially outperforms Logistic Regression (AUC 0.738 vs 0.559), confirming that the relationship between ESM2 features and binding is **non-linear**.
- Logistic Regression near AUC 0.5 indicates it cannot find a linear decision boundary in the 640-dimensional embedding space.
- AUC 0.738 is a strong baseline. Literature reports AUC 0.70–0.85 for similar setups with random negative generation.
- The gap between validation AUC (0.810) and test AUC (0.738) indicates mild overfitting; increasing data volume or regularisation would help.

Metric	Logistic Regression	Random Forest
AUC-ROC	0.559	0.738
AUPR	0.548	0.731
F1-score	0.490	0.743
Precision	0.498	0.722
Recall	0.481	0.683
Accuracy	0.498	0.714
Specificity	0.515	0.745

Table 1: Test-set performance of both baseline models. Bold = better model per metric.

3. Embedding Structure Analysis

UMAP projections of ESM2 embeddings were computed separately for TCR and epitope sequences to understand what structure the language model captures prior to any model training.

3.1. Key Findings

- **TCR-only UMAP:** binding and non-binding labels are fully mixed — consistent with biology (TCR sequence alone does not determine binding; the epitope context is required).
- **TCR+Epitope UMAP:** distinct isolated clusters emerge, several of which are strongly enriched for a single pathogen or MHC class. This confirms that the *joint* embedding captures binding-relevant structure.
- **Pathogen coloring:** isolated islands correspond almost entirely to one pathogen (e.g., all SARS-CoV-2 pairs in one region, all Influenza pairs in another).
- **MHC class:** MHC class I and class II pairs separate in embedding space, which is biologically expected since they present different peptide lengths and bind different TCR subtypes.

4. Feature Importance (SHAP)

SHAP (SHapley Additive exPlanations) was computed on the Random Forest to identify which of the 640 ESM2 dimensions contribute most to binding predictions.

- Both TCR and epitope embedding dimensions contribute; no single dimension dominates the predictions.
- Top SHAP features are distributed across both embedding halves, confirming that the model exploits both sequence modalities.
- High-SHAP dimensions correspond to latent biochemical properties (hydrophobicity, charge distribution) rather than identifiable sequence positions.

5. Limitations

- **Negative quality:** random mismatching may accidentally produce true positives, adding label noise to the training data.

- **Data leakage risk:** a random train/test split was used; an epitope-aware split (`GroupShuffleSplit`) would give a stricter and more realistic evaluation of generalisation to unseen epitopes.
- **Model size:** ESM2 8M is the smallest ESM2 variant; larger models (35M, 150M, 650M parameters) would produce richer representations.
- **Relational signal:** classical ML cannot capture the relational structure between TCRs and epitopes across patients and datasets.

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